Laser Cell 2 - Cut Quality Troubleshooting Guide

Internal working document, production engineering. Revision 3, March 2026.

Synthetic document written for a student project. It is not a TRUMPF manual; codes and values are illustrative.

1. How to use this guide

Inspect the cut edge first. Find the symptom in section 2, then check the listed causes from top to bottom; the most common cause is listed first. Check the machine message log (section 3) for warnings during the job. Record every maintenance action in the maintenance book.

2. Cut quality problems

Symptom	Possible causes (most likely first)	Remedy
Dross / burr on the bottom edge (stainless, aluminium, N2)	1. Cutting gas pressure too low or supply unstable 2. Nozzle worn or damaged (orifice no longer round) 3. Focus position too high 4. Feed rate too high	Check gas supply and message 1201/1202. Inspect nozzle under magnifier; replace if worn. Lower focus by 0.5 mm steps. Reduce feed 5-10 %.
Dross on mild steel (O2)	1. Feed rate too low (overburning) 2. Gas pressure too high 3. Material surface rusty or oily	Increase feed rate. Check pressure against cutting table. Use clean sheet.
Rough edge / heavy striations, lines bend backwards at the bottom	1. Feed rate too high for thickness 2. Worn nozzle 3. Contaminated protective glass (power loss)	Return to cutting-table feed rate. Replace nozzle. Check and clean protective glass.
Incomplete cut / parts not cut through	1. Contaminated or damaged protective glass (beam power reduced) 2. Feed rate too high 3. Focus position wrong 4. Laser power reduced (message 4001)	Clean or replace protective glass. Reduce feed. Run focus test. Contact service if 4001 persists after cleaning.
Burnt edges / melted corners (thick mild steel)	1. Beam quality degraded (dirty optics) 2. Feed too low in corners 3. Material too hot (dense nesting)	Clean optics. Check corner parameters. Increase part spacing.
Bevel angle out of tolerance	1. Bevel axis out of calibration (after collision or head crash) 2. Wrong bevel program or tool 3. Height sensor not calibrated (message 3302)	After any collision: calibrate height sensor and bevel axis before bevel jobs. Measure first part with protractor.

3. Machine messages

The machine writes every message to the message log with a timestamp. INFO messages are normal. WARNING messages need action before the next job. ERROR messages stop the program.

Code	Level	Text	Meaning / action
1201	WARNING	Cutting gas pressure below setpoint	Supply pressure too low. Check tank, regulator and lines. Expect dross on N2 cuts.
1202	INFO	Cutting gas pressure fluctuation	Short pressure drop. Repeated: check regulator.
2105	WARNING	Protective glass contamination - check and clean	Clean protective glass now. Dirty glass absorbs power and can crack.
3301	ERROR	Collision detected - cutting head	Program stops. Check nozzle, ceramic, head. Calibrate before restart.
3302	WARNING	Capacitive height sensor - calibration required	Calibrate sensor. Bevel cuts will be wrong until done.
4001	WARNING	Laser power reduced - beam path monitoring	Usually dirty optics. Clean glass; if it stays, call service.
5010	INFO	Program aborted	Program stopped by operator or by an error.
6002	WARNING	Sheet tip-up detected - check part	Tilted part may cause collision. Remove part.
7100 / 7200 / 7300	INFO	Door opened / Pallet change / Nozzle centring passed	Normal operation.
8000	INFO	Lubrication interval reminder	Lubricate at end of shift.

4. Maintenance intervals

Item	Interval	Note
Protective glass	Check every shift; clean when spatter visible	Replace if burnt spots or cracks
Nozzle	Inspect daily; replace every 40-60 operating hours or when damaged	N2 cutting is most sensitive to nozzle wear. Keep spare 2.3 mm and 2.7 mm nozzles.
Nozzle centring	After every nozzle change	Automatic check (7300)
Height sensor / bevel axis	After any collision; otherwise monthly	Service technician for bevel axis
Cutting gas supply	Weekly check of tank level and regulator	Facilities team

5. Cutting parameters

Use the cutting table (file cutting_table.csv) as the starting point. Do not increase feed rate above the table value by more than 10 % without a test cut. Programs ending in _FAST are not released by engineering.

6. Recording a quality problem

Mark the job NOK in the inspection sheet, write the defect type, measure dross height (mm) and roughness Ra (um) if a gauge is available, and take a photo of the cut edge named with the job ID.